

● HARD

● EXPRESSION OF IDEAS

Rhetorical Synthesis

10 questions · ~15 min

02

Q1

EXPRESSION OF IDEAS

While researching a topic, a student has taken the following notes:

- Nissologists are scientists who study islands.
- Some nissologists define an island as any piece of land surrounded by water.
- Using that definition, they determined that Sweden has 221,000 islands.
- Other nissologists define an island as being 1 kilometer square, a certain distance from the mainland, and having at least 50 permanent residents.
- Using that definition, they determined that Sweden has 24 islands.

The student wants to make and support a generalization about nissologists' definition of an island. Which choice most effectively uses relevant information from the notes to accomplish these goals?

- A. The definition of an island as any piece of land surrounded by water is supported by some nissologists, scientists who study islands.
- B. Multiple counts of Sweden's islands have been based on different definitions of an island.
- C. Based on a recent count, Sweden has a relatively small number of islands with at least 50 permanent residents.
- D. Nissologists' different definitions can result in huge disparities in counts of islands, as the example of Sweden shows.

While researching a topic, a student has taken the following notes:

- British scholar Robert Plot described fossilized dinosaur bones in his 1676 book *The Natural History of Oxfordshire*.
- Plot earned a reputation for being the first person to have discovered dinosaur remains.
- In 1990, archaeologists in Lesotho, in southern Africa, discovered a fossilized phalanx of a *Massospondylus carinatus* dinosaur in a cave once inhabited by humans.
- Indigenous Khoesan and Basotho peoples had inhabited the cave beginning around 1100 CE.
- According to paleontologist Julien Benoit, these peoples may have found the phalanx and brought it to the cave centuries before Plot's descriptions.

The student wants to emphasize the significance of the 1990 discovery to Plot's reputation. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Benoit challenged Plot's reputation for being the first person to have discovered *M. carinatus* remains.
- B. Evidence that Khoesan and Basotho peoples may have found an *M. carinatus* phalanx as long ago as 1100 CE suggests that Plot may not have been the first person to have discovered dinosaur remains.
- C. According to Benoit's analysis of the 1990 discovery, Indigenous peoples in southern Africa may have brought the fossilized phalanx to the cave as long ago as 1100 CE.

- D. In 1990, more than three centuries after Plot claimed in his book that he had found fossilized dinosaur bones, archaeologists uncovered evidence in southern Africa that disproved his claims.

While researching a topic, a student has taken the following notes:

- Modularity of mind is the notion that the mind is at least partly composed of innate neural structures (modules) that perform fast, necessary tasks.
- 1983: cognitive scientist Jerry A. Fodor hypothesized that low-level cognitive systems (e.g., perception, language) are modular.
- In Fodorian modularity, high-level systems (e.g., reasoning) are not modular.
- 2003: cognitive scientist Peter Carruthers proposed the massive modularity hypothesis (MMH).
- MMH expands modularity to include all cognitive systems.

The student wants to compare Fodor’s hypothesis with Carruthers’s. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. In considering some but not all cognitive systems modular, Fodorian modularity is not as expansive in its definition of modularity as MMH is.
- B. Following Fodor’s 1983 hypothesis, Carruthers proposed that modularity of mind includes all cognitive systems.
- C. The hypotheses of Fodor and Carruthers differ in whether they consider low-level cognitive systems, such as perception and language, modular.
- D. In 2003, Carruthers proposed the massive modularity hypothesis, disagreeing with Fodor’s earlier hypothesis that the mind is composed of innate neural structures.

While researching a topic, a student has taken the following notes:

- Neuroscientists Krishnan Padmanabhan and Zhen Chen sought to better understand the workings of the brain's olfactory system.
- They devised a study using mathematical models.
- They found that certain fibers allow the brain to toggle from one method of processing smells to another.
- In one method, cells in the piriform cortex (where the perception of odor forms) capture olfactory information at a given moment.
- In the other, the cells track changes in olfactory information over time.

The student wants to summarize the study's findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. To arrive at these findings, which describe dual methods of processing smells in the piriform cortex, Padmanabhan and Chen devised a study using mathematical models.
- B. Padmanabhan and Chen showed that olfactory information is captured by cells in the piriform cortex, where the perception of odor forms.
- C. Using mathematical models, Padmanabhan and Chen devised a study to better understand the workings of the brain's olfactory system.
- D. According to Padmanabhan and Chen, the brain can toggle between capturing olfactory information at a given moment and tracking changes in that information over time.

While researching a topic, a student has taken the following notes:

- In a 2023 study, environmental scientist Jazmin Locke-Rodriguez and colleagues tested the use of floating treatment wetlands (FTWs) in Florida.
- FTWs are artificial floating platforms of plants used to remediate polluted or nutrient-imbalanced water.
- Finding: FTWs using marigold flowers removed 52% more total phosphorus than the control.
- Finding: The test yielded 65 market-quality blooms per square meter.
- The authors concluded marigolds showed “promising potential as a commercially viable remediating crop cultivated on FTWs in South Florida.”

The student wants to present the findings of the study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The authors concluded that marigolds grown on FTWs were “commercially viable,” having produced 65 blooms per square meter of market-quality blooms in a 2023 study.
- B. In a 2023 study, Locke-Rodriguez and colleagues found that marigolds cultivated on FTWs produced 52% more market-quality flower blooms than the control.
- C. Locke-Rodriguez and colleagues found that FTWs using marigolds not only helped remove phosphorus from the water but also yielded market-quality blooms.
- D. FTWs using marigolds, Locke-Rodriguez and colleagues found, yielded 65 flower blooms and removed 52% of phosphorus from the water.

While researching a topic, a student has taken the following notes:

- The factors that affect clutch size (the number of eggs laid at one time) have been well studied in birds but not in lizards.
- A team led by Shai Meiri of Tel Aviv University investigated which factors influence lizard clutch size.
- Meiri's team obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.
- Larger clutch size was associated with environments in higher latitudes that have more seasonal change.
- Lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Researchers wanted to know which factors influence lizard egg clutch size because such factors have been well studied in birds but not in lizards.
- B. After they obtained data for over 3,900 lizard species, researchers determined that larger clutch size was associated with environments in higher latitudes that have more seasonal change.
- C. We now know that lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.
- D. Researchers obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.

While researching a topic, a student has taken the following notes:

- Archaeologist Jon Erlandson and colleagues argue that humans first arrived in the Americas by sea.
- They propose that humans traveled between Pacific Ocean islands and coastlines from northeast Asia to the Americas.
- Many of these islands and coastal zones were later submerged as glaciers melted and sea levels rose.
- The researchers think that “a coastal route, including kelp forests and estuaries, would have provided a rich mix of marine, estuarine, riverine, and terrestrial resources” such as seaweeds, fish, and birds.
- This proposed scenario is known as the kelp highway hypothesis.

The student wants to summarize the kelp highway hypothesis. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Pacific Ocean islands and coastlines likely contained “a rich mix of marine, estuarine, riverine, and terrestrial resources” such as seaweeds, fish, and birds, according to researchers.
- B. One argument about how humans first arrived in the Americas is the kelp highway hypothesis proposed by Jon Erlandson and colleagues.
- C. Humans may have first arrived in the Americas by sea, traveling between Pacific Ocean islands and coastlines and subsisting on a variety of resources.
- D. As glaciers melted and sea levels rose, many Pacific Ocean islands and coastal zones were submerged.

While researching a topic, a student has taken the following notes:

- Ulaanbaatar is the capital of Mongolia.
- The city's population is 907,802.
- Ulaanbaatar contains 31.98 percent of Mongolia's population.
- Hanoi is the capital of Vietnam.
- The city's population is 7,781,631.
- Hanoi contains 8.14 percent of Vietnam's population.

The student wants to emphasize the relative sizes of the two capitals' populations. Which choice most effectively uses information from the given sentences to emphasize the relative sizes of the two capitals' populations?

- A. Mongolia's capital is Ulaanbaatar, which has 907,802 people, and Vietnam's capital is Hanoi, which has 7,781,631 people.
- B. Comparing Vietnam and Mongolia, 7,781,631 is 8.14 percent of Vietnam's population, and 907,802 is 31.98 percent of Mongolia's.
- C. Even though Hanoi (population 7,781,631) is larger than Ulaanbaatar (population 907,802), Ulaanbaatar accounts for more of its country's population.
- D. The populations of the capitals of Mongolia and Vietnam are 907,802 (Ulaanbaatar) and 7,781,631 (Hanoi), respectively.

While researching a topic, a student has taken the following notes:

- In 1978, Sámi activists staged protests to block the construction of a dam on the Alta River in Norway.
- The dam would disrupt Sámi fishing and reindeer herding.
- The dam was ultimately built, but the Alta conflict had a lasting impact.
- It brought international attention to the issue of Sámi rights.
- It led to a set of 2005 legal protections establishing Sámi rights to lands, waters, and resources.

The student wants to make and support a generalization about the Alta conflict. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. During the Alta conflict, Sámi activists staged protests to block the construction of a dam on the Alta River in Norway that would disrupt local fishing and reindeer herding.
- B. Although the dam that the Sámi activists had protested was ultimately built, the Alta conflict had a lasting impact.
- C. Sámi rights to lands, waters, and resources received international attention and legal protections as a result of the Alta conflict.
- D. The Alta conflict had a lasting impact, resulting in international attention and legal protections for Sámi rights to lands, waters, and resources.

While researching a topic, a student has taken the following notes:

- In 1971, experimental musician Pauline Oliveros created *Sonic Meditations*.
- *Sonic Meditations* is not music but rather a series of sound-based exercises called meditations.
- Each meditation consists of instructions for participants to make, imagine, listen to, or remember sounds.
- The instructions for Meditation V state, “walk so silently that the bottoms of your feet become ears.”
- Those for Meditation XVIII state, “listen to a sound until you no longer recognize it.”

The student wants to provide an explanation and an example of Oliveros’s *Sonic Meditations*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. *Sonic Meditations* is not music but rather a series of sound-based meditations that consist of instructions; Meditation XVIII, for instance, instructs participants to “listen to a sound until you no longer recognize it.”
- B. In 1971, Oliveros created *Sonic Meditations*, a series of meditations that consist of instructions for participants to make, imagine, listen to, or remember sounds.
- C. “Walk so silently that the bottoms of your feet become ears” is one example of the instructions found in Oliveros’s *Sonic Meditations*.
- D. While both meditations consist of instructions, Meditation XVIII instructs participants to “listen,” whereas Meditation V instructs participants to “walk.”